DAY 1 AI ANSWERS AND MANUAL EXECUTION OUTPUT SCREENSHOTS

Linux File System Explorer

Description
Investigation

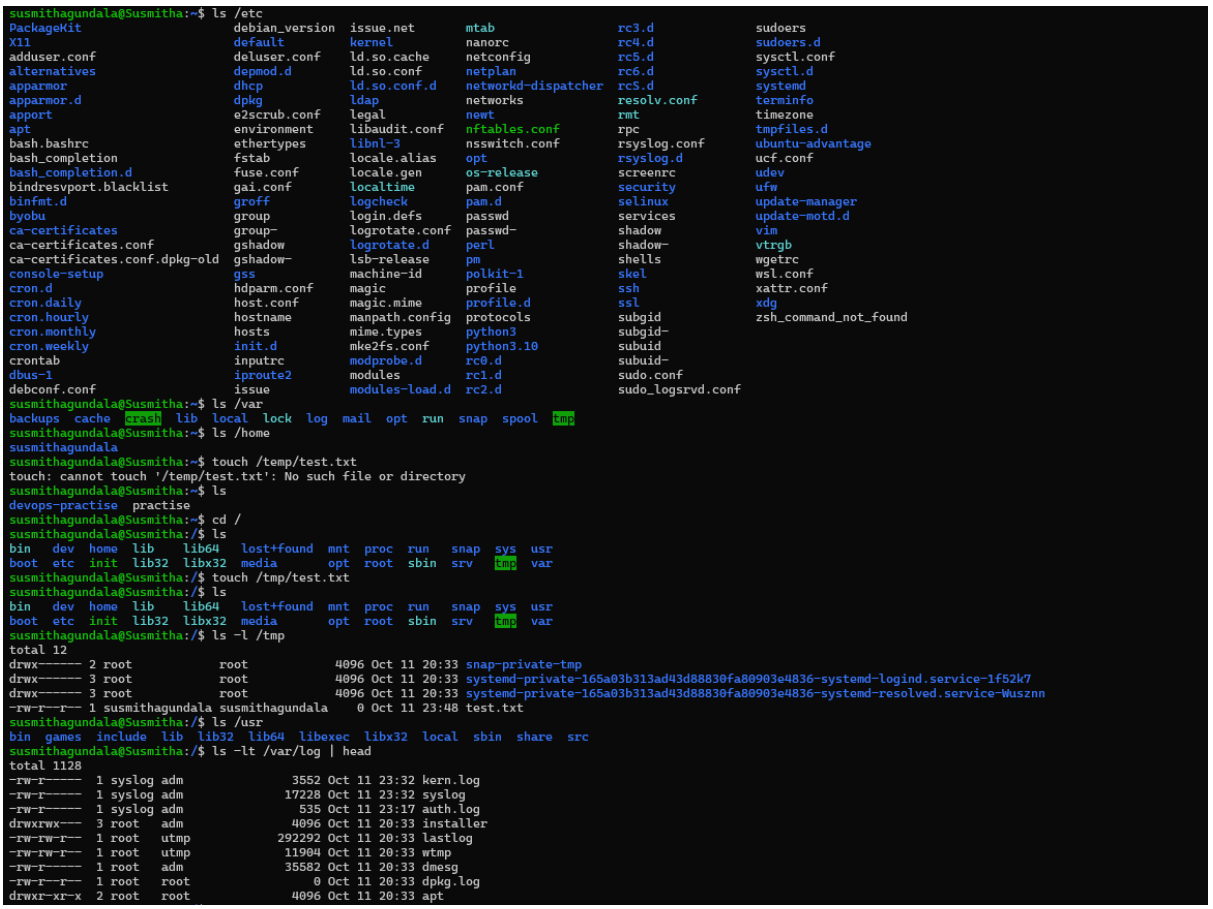
Ask AI:

1. **What is the purpose of /etc?** Stores system-wide configuration files that control how Linux and its services behave. Examples: /etc/hosts, /etc/passwd, /etc/ssh/ Used to configure networking, users, SSH, and other services.
2. **What is /var used for?** Stores data that changes frequently while the system runs. Examples: /var/log/, /var/cache/, /var/tmp/ Used for logs, caches, mail queues, and application data. Think of it as the system's changing records and activity folder.
3. **What is /home used for?** Stores personal files and settings for regular users. Example: /home/susmi/ Used for documents, downloads, projects, and user-specific configuration. Think of it as your personal workspace.
4. **What is /tmp used for?** Stores short-lived files created by applications and processes. Examples: temporary processing files and intermediate results. Files may be deleted automatically during reboot or by cleanup policies. Think of it as a scratchpad for temporary work.
5. **What is /usr used for?** Contains many installed applications, commands, libraries, and documentation. Despite its name, it is not primarily for personal user files. /usr/bin/ contains many everyday commands and programs. /usr/lib/ is libraries used by programs. /usr/share/ is shared data and documentation.

Challenge question

If an application is generating log files, which Linux directory would you normally investigate first, and why?

The answer is /var/log. By convention /var holds data that changes over time, and /var/log is the standard place for system and application logs.



Linux Commands Investigation

Description

AI questions

1. **What does pwd do?** pwd stands for Print Working Directory. It displays the full path of the directory you are currently in.
2. **Difference between ls and ls -la?** ls lists visible file names only. ls -la shows all files including hidden ones (-a) in long format (-l): permissions, owner, group, size, and modified date.
3. **What does cd do?** Changes the current directory. cd ~ goes to home directory, cd .. goes up one level, cd - returns to the previous directory, cd / moves to the root directory.
4. **What does mkdir do?** mkdir creates a new directory. mkdir -p a/b/c creates nested directories in one go.
5. **What does rm do?** rm removes (deletes) files. There is no recycle bin, so deletion is permanent. Use -r to remove directories recursively.

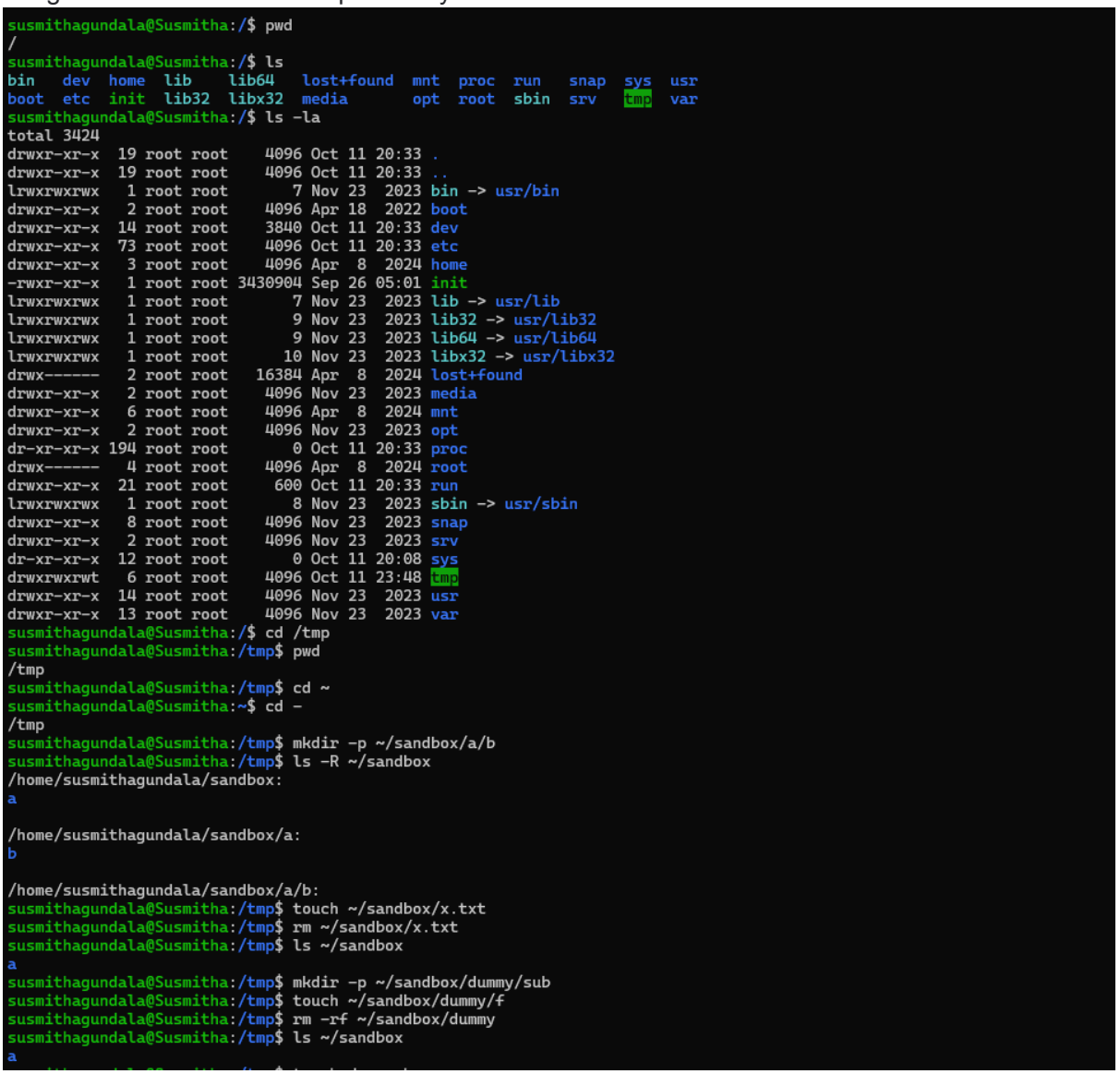
Challenge

Ask AI:

"What happens if I execute rm -rf?"

Then discuss why this command is dangerous. If I execute rm -rf, Linux deletes the specified files or folders and everything inside them without asking for confirmation in most cases. This command is dangerous because it can permanently delete important files, damage the operating

system, or cause data loss. Therefore, we should always check the file path carefully before using it and never run it on important system directories.



Linux Permissions

Description

AI questions

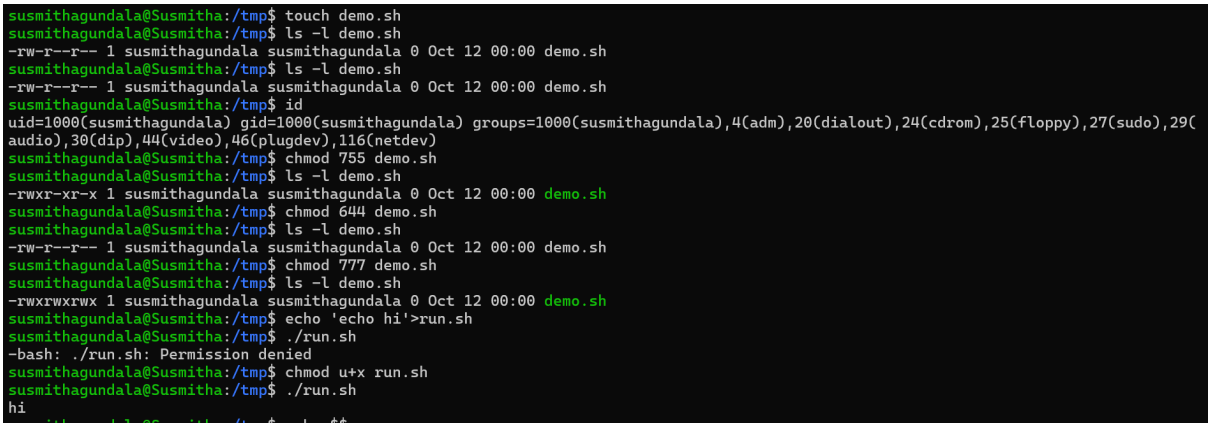
1. **What does rwx mean?** r = read (4), w = write (2), x = execute (1). For directories, x means you can enter/traverse it.
2. **What are owner, group and others?** Three permission classes: the file's owner (user), members of the file's group, and everyone else (others). In -rwxr-xr-- the three triplets are owner, group, others.
3. **What does chmod 755 mean?** Owner rwx (7), group r-x (5), others r-x (5). Common for scripts and directories: everyone can read and run, only the owner can modify.
4. **What does chmod 644 mean?** Owner rw- (6), group r-- (4), others r-- (4). Common for regular files: owner can edit, everyone else can only read.

5. **Why is chmod 777 generally discouraged?** It gives everyone read, write and execute. Any user or compromised process can modify or replace the file, which is a security risk, and it hides the real cause of the problem instead of fixing it.

Challenge

Ask:

Is chmod 777 always required when a user gets "Permission denied"? No, almost never. Check ls -l and the parent directories (x needed on each), identify who needs what access, then grant the minimum (chmod u+x, chmod 640, chown user:group). Other causes: wrong user, sudo needed, SELinux/AppArmor, or a read-only filesystem.



Linux Processes

Description

This introduces a little troubleshooting thinking.

AI questions

1. **What is a Linux process?**
2. **What does ps show?**
3. **What does ps aux show?**
4. **What does top do?**
5. **Why would a developer investigate processes?**

Challenge

Ask AI:

"If an application is running slowly, can top help investigate the problem?"

Then explain what information they can actually obtain from it. No, almost never. Check ls -l and the parent directories (x needed on each), identify who needs what access, then grant the minimum (chmod u+x, chmod 640, chown user:group). Other causes: wrong user, sudo needed, SELinux/AppArmor, or a read-only filesystem.

```
susmithagundala@Susmitha:/tmp$ top
top - 00:07:15 up  2:56,  1 user,  load average: 0.24, 0.12, 0.
Tasks:  33 total,   2 running,  31 sleeping,   0 stopped,   0 z
%Cpu(s):  5.5 us,  7.1 sy,  0.0 ni, 85.0 id,  0.0 wa,  0.0 hi,
MiB Mem :   3806.4 total,   3115.2 free,    508.1 used,    183.
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3161.

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
  36492 susmith+  20   0    3212   2020   1900 R 100.0   0.1
      1 root      20   0  167064  12784   8448 S   6.7   0.3
      2 root      20   0    3760   2332   2204 S   0.0   0.1
      3 root      20   0    3612   2100   2004 S   0.0   0.1
      8 root      20   0    3760   2104   2004 S   0.0   0.1
     41 root      19  -1   47820  15480  14468 S   0.0   0.4
     62 root      20   0   22888   6380   4832 S   0.0   0.2
     72 root      20   0    4764   1676   1148 S   0.0   0.0
     73 root      20   0    4504   1476   1276 S   0.0   0.0
     79 root      20   0    4720   1748   1264 S   0.0   0.0
     80 root      20   0    4636    968    760 S   0.0   0.0
     83 root      20   0    4968   1748   1252 S   0.0   0.0
     90 systemd+  20   0   25540  13740   9488 S   0.0   0.4
    117 root      20   0    4312   2868   2616 S   0.0   0.1
    119 message+  20   0    8600   4900   4240 S   0.0   0.1
    130 root      20   0   30104  19624  10836 S   0.0   0.5
    131 syslog    20   0  222408   5952   4600 S   0.0   0.2
    133 root      20   0 1763044  31588  19652 S   0.0   0.8
    134 root      20   0   15332   7868   6880 S   0.0   0.2
    222 root      20   0  107236  22120  13880 S   0.0   0.6
    229 root      20   0    3200   2280   2144 S   0.0   0.1
    347 root      20   0    3776   1304   1176 S   0.0   0.0
    348 root      20   0    3844   1436   1240 S   0.0   0.0
    349 susmith+  20   0    6184   5484   3680 S   0.0   0.1
    350 root      20   0    7536   5156   4232 S   0.0   0.1
    379 susmith+  20   0   17068   9724   8104 S   0.0   0.2
    380 susmith+  20   0  103276   5096   1684 S   0.0   0.1
    392 susmith+  20   0    6076   5220   3532 S   0.0   0.1
susmithagundala@Susmitha:/tmp$ kill %1
susmithagundala@Susmitha:/tmp$
```

```
susmithagundala@Susmitha:/tmp$ top
top - 00:06:29 up  2:55,  1 user,  load average: 0.05, 0.08, 0.
Tasks:  34 total,   1 running,  33 sleeping,   0 stopped,   0 z
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,
MiB Mem :   3806.4 total,   3115.9 free,    506.5 used,    184.
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3163.

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
      1 root      20   0  167064  12784   8448 S   0.7   0.3
    685 root      20   0  4m236  37568   9952 D   0.3   1.0
      2 root      20   0    3760   2332   2204 S   0.0   0.1
      3 root      20   0    3612   2100   2004 S   0.0   0.1
      8 root      20   0    3760   2104   2004 S   0.0   0.1
     41 root      19  -1   47820  15480  14468 S   0.0   0.4
     62 root      20   0   22888   6388   4832 S   0.0   0.2
     72 root      20   0    4764   1676   1148 S   0.0   0.0
     73 root      20   0    4504   1476   1276 S   0.0   0.0
     79 root      20   0    4720   1748   1264 S   0.0   0.0
     80 root      20   0    4636    968    760 S   0.0   0.0
     83 root      20   0    4968   1748   1252 S   0.0   0.0
     90 systemd+  20   0   25540  13740   9488 S   0.0   0.4
    117 root      20   0    4312   2868   2616 S   0.0   0.1
    119 message+  20   0    8600   4908   4240 S   0.0   0.1
    130 root      20   0   30104  19624  10836 S   0.0   0.5
    131 syslog    20   0  222408   5952   4600 S   0.0   0.2
    133 root      20   0 1763044  31588  19652 S   0.0   0.8
    134 root      20   0   15332   7868   6880 S   0.0   0.2
    222 root      20   0  107236  22120  13880 S   0.0   0.6
    229 root      20   0    3200   2280   2144 S   0.0   0.1
    347 root      20   0    3776   1304   1176 S   0.0   0.0
    348 root      20   0    3844   1436   1240 S   0.0   0.0
    349 susmith+  20   0    6184   5484   3680 S   0.0   0.1
    350 root      20   0    7536   5156   4232 S   0.0   0.1
    379 susmith+  20   0   17068   9724   8104 S   0.0   0.2
    380 susmith+  20   0  103276   5096   1684 S   0.0   0.1
    392 susmith+  20   0    6076   5220   3532 S   0.0   0.1
susmithagundala@Susmitha:/tmp$ ps aux --sort=-%cpu | head -5
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  1.2  0.3 167064 12784 ?        Ss   Oct11   1:50 /sbin/init
root         685  0.6  0.9 4m4236 37568 ?        S    Oct11   0:54 python3 /snap/ubuntu-desktop-installer/1286/usr/bin/cloud-init status --wait
root         681  0.1  2.0 943024 78532 ?        Sl   Oct11   0:10 /snap/ubuntu-desktop-installer/1286/usr/bin/python3.10 -m subiquity.cmd.server --use-os-prober --storage-version=2 --postinst-hooks-dir=/snap/ubuntu-desktop-installer/1286/etc/subiquity/postinst.d
root           2  0.0  0.0      0     0 ?        S    Oct11   0:00 [kthreadd]
susmithagundala@Susmitha:/tmp$ yes > /dev/null &
[1] 36492
```

```
susmithagundala@Susmitha:/tmp$ echo $$
349
susmithagundala@Susmitha:/tmp$ ps -p $$
    PID TTY          TIME CMD
    349 pts/0    00:00:00 bash
susmithagundala@Susmitha:/tmp$ ps
    PID TTY          TIME CMD
    349 pts/0    00:00:00 bash
  36140 pts/0    00:00:00 ps
susmithagundala@Susmitha:/tmp$ ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  1.2  0.3 167064 12784 ?        Ss   Oct11   1:49 /sbin/init
root           2  0.0  0.0   3760  2332 hvc0     Sl+  Oct11   0:00 /init
root           3  0.0  0.0   3612  2100 hvc0     S+   Oct11   0:00 init-watcher
root           8  0.0  0.0   3760  2104 hvc0     Sl+  Oct11   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          41  0.0  0.3  47820 15480 ?        S<s  Oct11   0:00 /lib/systemd/systemd-journald
root          62  0.0  0.1  22888  6380 ?        Ss   Oct11   0:02 /lib/systemd/systemd-udevd
root          72  0.0  0.0   4764  1676 ?        Ss   Oct11   0:01 snapfuse /var/lib/snapd/snaps/core22_864.snap /snap/core22/864 -o ro,nodev,allow_other,suid
root          73  0.0  0.0   4504  1476 ?        Ss   Oct11   0:00 snapfuse /var/lib/snapd/snaps/bare_5.snap /snap/bare/5 -o ro,nodev,allow_other,suid
root          79  0.0  0.0   4720  1748 ?        Ss   Oct11   0:02 snapfuse /var/lib/snapd/snaps/snapd_20290.snap /snap/snapd/20290 -o ro,nodev,allow_other,suid
susmithagundala@Susmitha:/tmp$ top
top - 00:06:07 up  2:55,  1 user,  load average: 0.07, 0.08, 0.
Tasks:  34 total,   1 running,  33 sleeping,   0 stopped,   0 z
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,
MiB Mem :   3806.4 total,   3116.7 free,    505.7 used,    184.
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3164.

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
      1 root      20   0  167064  12784   8448 S   0.7   0.3
    681 root      20   0  943024  78532  22764 S   0.3   2.0
    685 root      20   0   44236  37568   9952 S   0.3   1.0
      2 root      20   0    3760   2332   2204 S   0.0   0.1
      3 root      20   0    3612   2100   2004 S   0.0   0.1
      8 root      20   0    3760   2104   2004 S   0.0   0.1
     41 root      19  -1   47820  15480  14468 S   0.0   0.4
     62 root      20   0   22888   6380   4832 S   0.0   0.2
     72 root      20   0    4764   1676   1148 S   0.0   0.0
     73 root      20   0    4504   1476   1276 S   0.0   0.0
     79 root      20   0    4720   1748   1264 S   0.0   0.0
     80 root      20   0    4636    968    760 S   0.0   0.0
     83 root      20   0    4968   1748   1252 S   0.0   0.0
     90 systemd+  20   0   25540  13740   9488 S   0.0   0.4
    117 root      20   0    4312   2868   2616 S   0.0   0.1
    119 message+  20   0    8600   4900   4240 S   0.0   0.1
    130 root      20   0   30104  19624  10836 S   0.0   0.5
    131 syslog    20   0  222408   5952   4600 S   0.0   0.2
    133 root      20   0 1763044  31588  19652 S   0.0   0.8
    134 root      20   0   15332   7868   6880 S   0.0   0.2
    222 root      20   0  107236  22120  13880 S   0.0   0.6
    229 root      20   0    3200   2280   2144 S   0.0   0.1
    347 root      20   0    3776   1304   1176 S   0.0   0.0
    348 root      20   0    3844   1436   1240 S   0.0   0.0
    349 susmith+  20   0    6184   5484   3680 S   0.0   0.1
    350 root      20   0    7536   5156   4232 S   0.0   0.1
    379 susmith+  20   0   17068   9724   8104 S   0.0   0.2
    380 susmith+  20   0  103276   5096   1684 S   0.0   0.1
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